CHM-110 | WEEK 10 EXAM CHEAT SHEET

Recognize the clue ⟶ choose the equation ⟶ cancel units ⟶ answer. All 14 quiz questions.

01 Gas stoichiometry (not STP) — 23.8 L

IDENTIFY: Grams of *one* reactant → liters of a *different* gas; P, T given.

FACTOR-LABEL SETUP:

$$6.29\,\text{g C}_4\text{H}_{10} \times \frac{1\,\text{mol C}_4\text{H}_{10}}{58.12\,\text{g C}_4\text{H}_{10}} \times \frac{10\,\text{mol H}_2\text{O}}{2\,\text{mol C}_4\text{H}_{10}}$$

$$V = 0.5411\,\text{mol} \times \frac{0.08206\,\text{L}\cdot\text{atm}}{\text{mol}\cdot\text{K}} \times \frac{600.\,\text{K}}{850\,\text{Torr} \times \frac{1\,\text{atm}}{760\,\text{Torr}}}$$

02 Gas stoichiometry at STP — 28.8 L

IDENTIFY: Grams of one substance → gas liters; explicitly says **STP**.

FACTOR-LABEL SETUP:

$$59.2\,\text{g Na} \times \frac{1\,\text{mol Na}}{22.99\,\text{g Na}} \times \frac{1\,\text{mol H}_2}{2\,\text{mol Na}}$$

$$\times \frac{22.4\,\text{L H}_2}{1\,\text{mol H}_2} = 28.8\,\text{L H}_2$$

03 Enthalpy sign — Exothermic

IDENTIFY: ΔH is given; identify whether energy is absorbed or released.

FACTOR-LABEL SETUP:

$\Delta H = -73.54\,\text{kJ} < 0 \quad \Longrightarrow \quad$ heat released

$\Delta H < 0$: EXOthermic $\qquad \Delta H > 0$: ENDOthermic

No numerical calculation needed. Negative enthalpy means heat exits the system.

04 Calorimetry: find mass — 2.23 kg

IDENTIFY: Heat q, specific heat c, and two temperatures; find mass.

FACTOR-LABEL SETUP:

$$m = \frac{q}{c\,(T_f - T_i)}$$

$$\frac{5310\,\text{J}}{\left(0.130\,\frac{\text{J}}{\text{g}\cdot^\circ\text{C}}\right)(38.3 - 20.0)\,^\circ\text{C}} \times \frac{1\,\text{kg}}{1000\,\text{g}}$$

05 Molar enthalpy of solution — +25.7 kJ/mol

IDENTIFY: Mass of solute plus kJ; choices expressed in kJ/mol.

FACTOR-LABEL SETUP:

$$\Delta H_{\text{mol}} = \frac{0.280\,\text{kJ}}{0.873\,\text{g}} \times \frac{80.052\,\text{g}}{1\,\text{mol}}$$

$= +25.7\,\text{kJ/mol} \quad \Longrightarrow \quad$ heat absorbed

On this multiple-choice question, the positive ΔH alone identifies the answer.

06 Concepts / energy diagrams — Conceptual

IDENTIFY: Energy in reactants or products; warm/cold; energy-level graph.

FACTOR-LABEL SETUP:

$\Delta H = H_{\text{products}} - H_{\text{reactants}}$

Products higher: endo $(+\Delta H)$; lower: exo $(-\Delta H)$

Heat on left ⇒ endo; heat on right ⇒ exo.

Specific heat c: heat needed for 1 g to rise by 1 °C.

07 Gas liters → solid grams — 173 g

IDENTIFY: Gas P, V, T known; mass of a *different* reactant requested.

FACTOR-LABEL SETUP:

$$n_{\text{CO}_2} = \frac{(5.30\,\text{atm})(25.0\,\text{L})}{\left(0.08206\,\frac{\text{L}\cdot\text{atm}}{\text{mol}\cdot\text{K}}\right)(1523.15\,\text{K})}$$

$$1.060\,\text{mol CO}_2 \times \frac{1\,\text{mol Pb(CO}_3)_2}{2\,\text{mol CO}_2} \times \frac{327.22\,\text{g Pb(CO}_3)_2}{1\,\text{mol Pb(CO}_3)_2}$$

EXAM Choose the shortest path — Don't overcalculate

IDENTIFY: Identify units of what is *given* and what is *requested*.

FACTOR-LABEL SETUP:

g A → mol A $\xrightarrow{\text{coefficients}}$ mol B → g or L B

At STP: 22.4 L/mol; otherwise: $PV = nRT$

For energy questions: kJ ↔ mol via ΔH, then mol ↔ g via molar mass.

EXAM CONSTANTS & CONVERSIONS 1 mol gas ≈ 22.4 L at STP (0 °C, 1 atm) · 760 Torr = 1 atm · T(K) = T(°C) + 273.15 · R = 0.08206 L atm/(mol K)

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08 Enthalpy sign — Exothermic

IDENTIFY: Given $\Delta H = -54.74\,\text{kJ}$; reaction type requested.

FACTOR-LABEL SETUP:

$$\Delta H = -54.74\,\text{kJ} < 0 \quad \Longrightarrow \quad \text{EXOthermic}$$

$$\text{Exothermic} = \text{energy released into surroundings}$$

Same recognition rule as Q03. No calculation needed.

09 Ideal Gas Law: pressure — 2.28 atm

IDENTIFY: Given moles n, volume V, temperature T; pressure unknown.

FACTOR-LABEL SETUP:

$$P = \frac{nRT}{V}$$

$$P = \frac{(0.545\,\text{mol})\left(0.08206\,\frac{\text{L·atm}}{\text{mol·K}}\right)(22.0 + 273.15)\,\text{K}}{5.80\,\text{L}}$$

10 Thermochemistry: kJ → grams — 357 g

IDENTIFY: Reaction has ΔH in kJ; energy given; grams consumed requested.

FACTOR-LABEL SETUP:

$$936\,\text{kJ} \times \frac{2\,\text{mol}\,HC_2H_3O_2}{315\,\text{kJ}} \times \frac{60.052\,\text{g}\,HC_2H_3O_2}{1\,\text{mol}\,HC_2H_3O_2} = 357\,\text{g}$$

11 Molecules → gas liters — 395 L

IDENTIFY: Molecules (not mol) given, plus P, T; volume requested.

FACTOR-LABEL SETUP:

$$7.32 \times 10^{24}\,\text{molecules} \times \frac{1\,\text{mol}}{6.022 \times 10^{23}\,\text{molecules}}$$

$$V = 12.155\,\text{mol} \times \frac{0.08206\,\text{L}\cdot\text{atm}}{\text{mol}\cdot\text{K}} \times \frac{400.\,\text{K}}{768\,\text{Torr} \times \frac{1\,\text{atm}}{760\,\text{Torr}}}$$

12 Ideal Gas Law: find grams — 6.46 g

IDENTIFY: Gas P, V, T supplied; mass requested; *no* reaction. 530. mL = 0.530 L; 28.0°C = 301.15 K.

FACTOR-LABEL SETUP:

$$n = \frac{(3.96\,\text{atm})(0.530\,\text{L})}{\left(0.08206\,\frac{\text{L·atm}}{\text{mol·K}}\right)(301.15\,\text{K})}$$

$$m = 0.08493\,\text{mol}\,N_2O_3 \times \frac{76.02\,\text{g}\,N_2O_3}{1\,\text{mol}\,N_2O_3}$$

13 STP gas liters → kJ — 114 kJ released

IDENTIFY: Gas volume at STP; reaction has kJ on product side.

FACTOR-LABEL SETUP:

$$83.2\,\text{L}\,H_2 \times \frac{1\,\text{mol}\,H_2}{22.4\,\text{L}\,H_2} \times \frac{92.0\,\text{kJ}}{3\,\text{mol}\,H_2}$$

$$= 114\,\text{kJ released} \qquad (\Delta H < 0)$$

The supplied equation omitted 2 before NH_3; the 3 mol H_2 : 92.0 kJ ratio is unchanged.

14 Calorimetry: heat q — 5500 cal released

IDENTIFY: Mass and specific heat given; substance heats or cools.

FACTOR-LABEL SETUP:

$$q = mc\,(T_f - T_i)$$

$$(100.\,\text{g})\left(1\,\frac{\text{cal}}{\text{g}\cdot{}^\circ\text{C}}\right)(25 - 80)\,{}^\circ\text{C}$$

$$= -5500\,\text{cal} \quad \Longrightarrow \quad 5500\ \text{cal released}$$

EXAM Dimensional analysis rules — Unit cancellation

IDENTIFY: Every unwanted unit should appear in a numerator and denominator.

FACTOR-LABEL SETUP:

$$\frac{\text{mol}}{\text{g}} \times \frac{\text{mol B}}{\text{mol A}} \times \frac{\text{L}}{\text{mol}} \quad \text{(as needed)}$$

$$q = mc\Delta T \quad \text{or} \quad \frac{\text{kJ}}{\text{mol reaction}}$$

Retain unrounded values until the end. Round to the given measurements' significant figures.

EXAM CONSTANTS & CONVERSIONS 1 L = 1000 mL · 1 kg = 1000 g · 1 mol = 6.022×10^{23} particles · $q < 0$ when cooling · $+\Delta H$ endothermic (absorbed), $-\Delta H$ exothermic (released)